

Understanding Patterns of Violent Crime Across Cities

Introduction & Context

Violent crime is a complex social issue shaped by multiple interacting factors, including offense type, time, demographic characteristics, and geographic context. Understanding these patterns is essential not only for public safety officials and policymakers, but also for community members seeking to better understand the risks and dynamics within their cities.

This data story explores reported violent crime incidents across cities in Maryland using incident-level data from the FBI's National Incident-Based Reporting System (NIBRS). Through a sequence of interactive dashboards, the project examines what types of violent crime are most common, how crime patterns change over time, who is most affected by violent crime, who is most likely to commit it, and where incidents tend to concentrate geographically. The goal of this project is to convey meaningful, data-driven insights that help audiences better understand violent crime patterns and consider how such insights could inform prevention and resource allocation strategies.

Research Questions

This data story is guided by four main research questions:

1. Which types of violent crime occur most frequently?
2. How do violent crime patterns change over time?
3. Who is most affected by violent crime?
4. Where do violent crime incidents concentrate geographically?

Together, these questions move from a high-level overview to more detailed demographic and spatial analysis, forming a coherent narrative across the visualization.

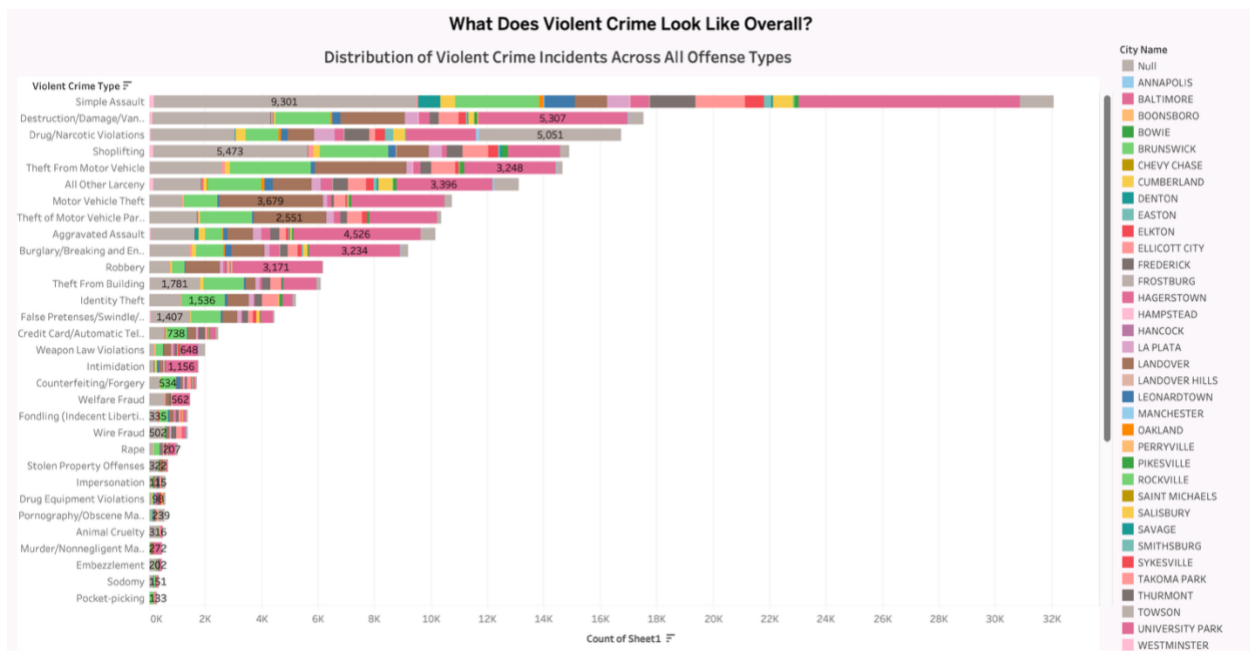
Intended Audience

This visualization is designed for a general, non-expert audience. While the data originates from official crime reporting systems, the dashboards and story emphasize clarity, readability, and narrative flow rather than technical complexity. The project is intended for viewers who may not have a background in data analysis but are interested in understanding crime trends and patterns that affect communities.

Overview of the Visualization

The data story is presented as a single Tableau Story composed of multiple dashboards, each representing a key step in the analytical narrative. The story begins with an overview of violent crime distribution across offense types, then moves through temporal patterns, victim and offender demographics, and finally geographic concentration. Each dashboard is accompanied by concise captions that guide interpretation and connect individual views to the broader narrative.

The overview dashboard provides a high-level summary of all reported violent crime incidents across offense types. Rather than limiting the view to only the most common categories, this dashboard displays the full distribution, revealing a long-tailed structure in which a small number of offense types account for a large share of incidents, while many others occur far less frequently. This context establishes the foundation for the analyses that follow.



Temporal Patterns: When Do Violent Crimes Occur?

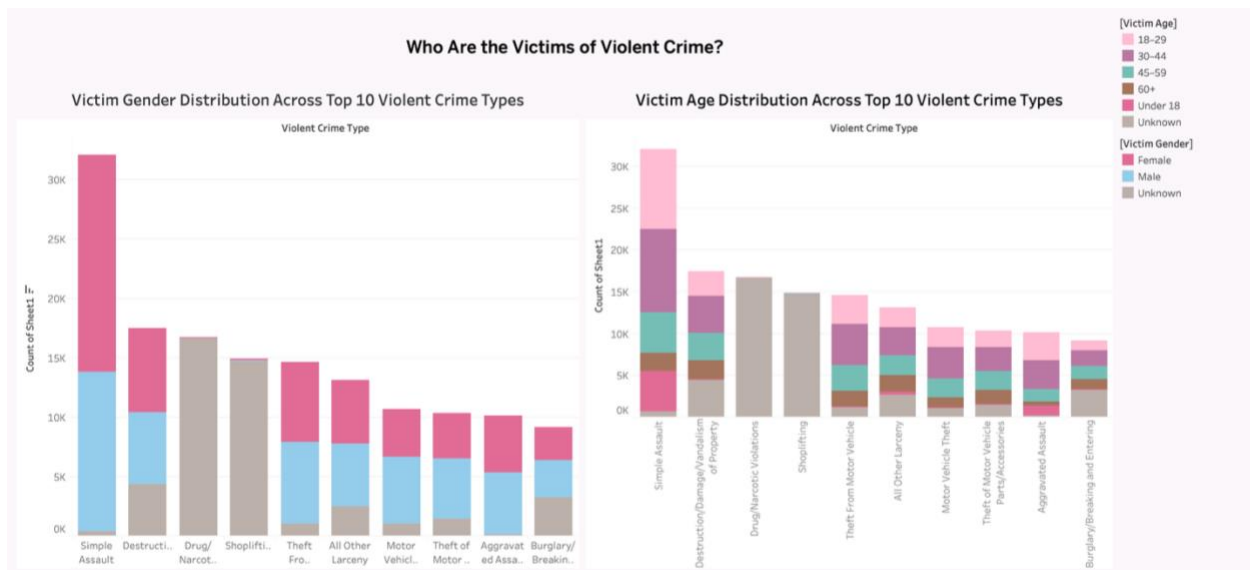
To understand when violent crimes occur most often, the next dashboard examines patterns across months and days of the week. A line chart shows monthly fluctuations in violent crime incidents, revealing relatively stable overall levels with modest seasonal variation. Certain months exhibit slightly higher incident counts, suggesting potential connections to seasonal factors such as weather, holidays, or changes in routine activities.



Complementing this view, a stacked bar chart displays incidents by day of the week. Crime is not evenly distributed across the week; instead, certain weekdays show consistently higher counts. These patterns suggest that violent crime is influenced by routine daily and weekly cycles related to work schedules, social activity, and public behavior rather than occurring randomly over time.

Victim Demographics: Who Is Most Affected?

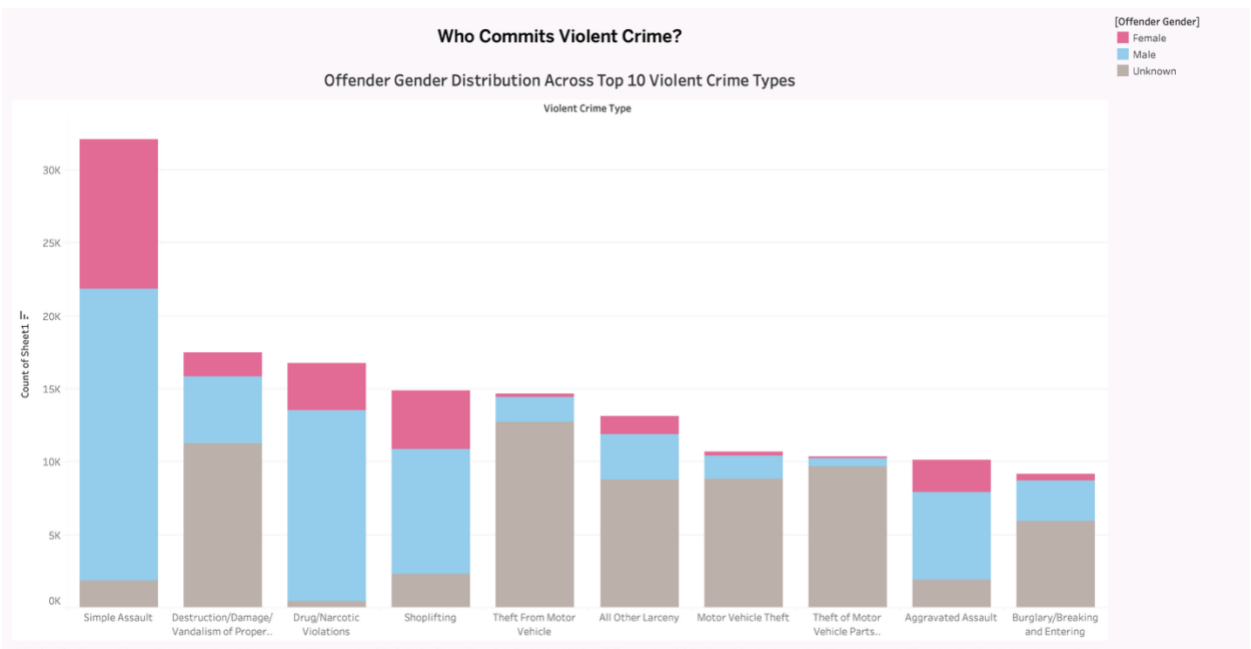
The next section focuses on victim characteristics, examining how violent crime impacts different demographic groups. Stacked bar charts show victim gender and age distributions across the top ten most common violent crime types. The visualizations reveal clear gender-based differences: female victims are disproportionately represented in offenses such as simple assault and domestic-related incidents, while male victims appear more frequently in drug-related and property-related violent crimes.



Age-based patterns further highlight vulnerability differences. Young adults aged 18-29 consistently represent the largest share of victims across many high-volume offense types, including simple assault and robbery. In contrast, certain offenses such as fraud and identity theft show a broader age distribution with greater representation among older age groups. These findings suggest that life stage, routine activities, and social exposure play important roles in shaping victimization risk.

Offender Demographics: Who Commits Violent Crime?

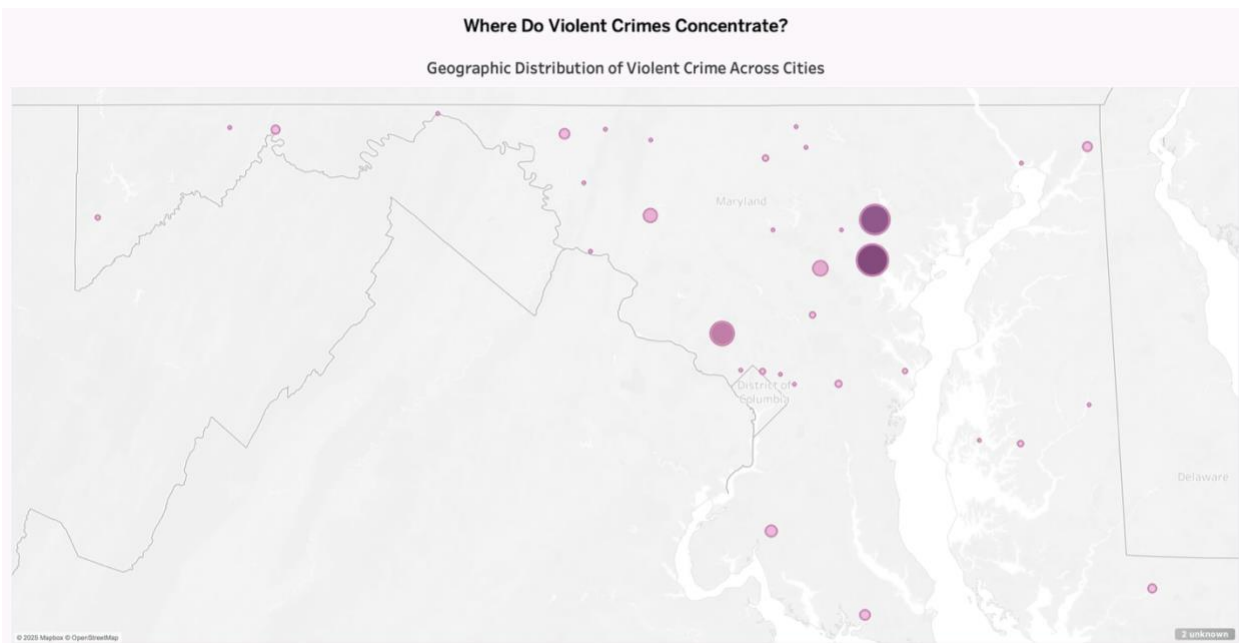
To complement the victim analysis, the following dashboard examines offender gender patterns across the same top ten violent crime categories. The visualization reveals a strong concentration of male offenders across nearly all offense types, particularly in robbery, burglary, and aggravated assault. Female offenders appear far less frequently and are more visible only in a small subset of crime categories.



Comparing offender and victim patterns side by side highlights an important distinction between who is most affected by violent crime and who is most likely to commit it. This contrast adds depth to the narrative and emphasizes the value of examining multiple perspectives within crime data.

Geographic Patterns: Where Do Violent Crimes Concentrate?

The final analytical dashboard visualizes the geographic distribution of violent crime incidents across cities using a proportional symbol map. Circle size and color intensity represent the total number of reported incidents in each city. The map reveals clear clustering in major urban centers, particularly around large metropolitan areas, while surrounding and rural areas show substantially lower incident density.



By retaining all crime types rather than filtering to only the most common offenses, this geographic view preserves spatial context and avoids oversimplification. The observed clustering highlights the role of population density, urban environments, and social infrastructure in shaping violent crime exposure.

Data and Design Decisions

Several design choices were made intentionally to support clarity and storytelling. Bar charts and stacked bars were used to facilitate comparison across categories, while line charts highlighted trends over time. The geographic map uses proportional symbols to emphasize relative differences in incident volume across cities without obscuring spatial relationships.

One limitation of the data is that it relies on reported incidents, which may underestimate true crime levels due to underreporting. Additionally, the analysis is descriptive rather than causal; while patterns are identified, the visualizations do not establish direct explanations for why these patterns occur. Some demographic fields also include “unknown” values, which may affect the interpretation of certain distributions.

Next Steps

Future work could extend this project in several directions. Incorporating multiple years of data would allow for analysis of long-term trends and changes over time. Integrating socioeconomic indicators such as income, unemployment, or housing density could provide deeper insight into structural factors associated with violent crime. Finally, more granular geographic analysis within urban areas could help identify neighborhood-level patterns that are not visible at the city level.

Conclusion

By organizing multiple dashboards into a single, cohesive data story, this project demonstrates how visualization can transform complex crime data into an accessible narrative. The findings highlight that violent crime is not evenly distributed across offense types, time, demographics, or geography. Understanding these patterns is a crucial step toward informed discussion, targeted prevention strategies, and data-driven decision-making.